

CFO Estimation for OFDM-Based Wireless Communication Systems Under Impulsive Noise

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March 13 , 2026

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Motivation and Problem Statement

The Impact of CFO

Modern wireless standards — **4G LTE, 5G NR, Wi-Fi (802.11ax)** — all rely on orthogonal frequency division multiplexing (OFDM) for its high spectral efficiency and resilience to multipath fading. However, OFDM is highly sensitive to synchronization errors: carrier frequency offset (CFO), caused by oscillator mismatch or Doppler shifts, destroys subcarrier orthogonality and leads to severe inter-carrier interference (ICI), significantly degrading bit error rate (BER) performance.

Limitations of Existing DL Estimators

Most deep learning (DL) approaches for CFO estimation rely on the conventional minimum mean squared error (MMSE) criterion.

- MMSE minimizes average squared error, relying on second-order statistics.
- Under non-Gaussian or impulsive noise, squared-error terms **amplify** high-amplitude outliers.
- This causes **gradient explosion** during training, hindering convergence.

System Model Overview

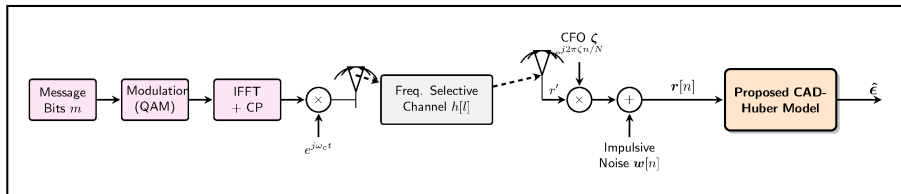


Figure: Signal propagation: transmitter, channel, CFO, impulsive noise, and CAD-Huber estimator.

Impulsive Noise Model

Symmetric α -Stable ($S\alpha S$) Distribution

We model $w[n]$ using the $S\alpha S$ distribution to capture impulsive industrial/urban channels. Its characteristic function is:

$$\varphi(t) = \exp(-\gamma |t|^\alpha)$$

Parameters

- $\alpha \in (0, 2]$: *characteristic exponent*
— controls outlier severity.
- $\gamma > 0$: *dispersion parameter*.

Key Implication

For $\alpha < 2$, variance is **infinite**.
Conventional L_2 -norm estimators diverge
— necessitating a robust alternative.

CAD-HuberNet Architecture

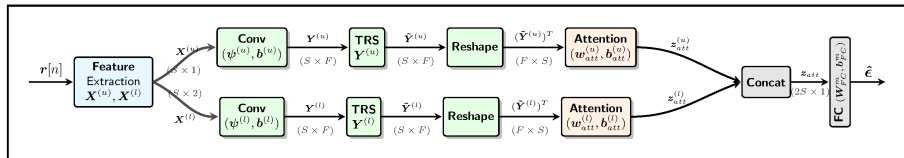


Figure: Detailed view of the dual-stream CNN-Attention-DNN (CAD-Huber) model.

Multi-Modal Input Feature Representation

Dual-Stream Inputs: $\mathbf{X} = [X^{(u)}, X^{(l)}]$

Upper Stream (Squared Spectrum)

Amplifies cyclic frequencies linked to carrier offset:

$$X^{(u)} = |\mathcal{F}(|r[n]|^2)| \in \mathbb{R}^{S \times 1}$$

Lower Stream (Magnitude + Phase)

Preserves phase information to resolve frequency ambiguities:

$$X^{(l)} = [|\mathcal{F}(r[n])|, \angle \mathcal{F}(r[n])] \in \mathbb{R}^{S \times 2}$$

Raw time-domain signals lack direct spectral features; the dual-stream design ensures both power-spectral and phase information are fed to the network simultaneously.

CAD-HuberNet: Key Components

1. Feature Extraction — Triple Residual Stacks (TRS)

Prevents vanishing gradients via skip connections. For the l -th layer:

$$Z_l = \sigma(W_l * Z_{l-1} + b_l) + Z_{l-1}$$

2. Attention Mechanism

Prioritizes spectral features from clean subcarriers; suppresses impulsive-spike-corrupted ones:

$$e_i = v^T \tanh(W_a h_i + b_a), \quad \alpha_i = \frac{\exp(e_i)}{\sum_k \exp(e_k)}$$

3. DNN Regression Head

Attention-weighted context vectors from both streams are concatenated and passed through fully-connected layers to produce the scalar CFO estimate $\hat{\epsilon}$.

Robust Optimization via Huber Loss

The Huber Penalty Function

To mitigate exploding gradients from heavy-tailed $S\alpha S$ noise, the network is trained with the Huber loss:

$$\rho_{\delta}(e) = \begin{cases} \frac{1}{2}e^2 & |e| \leq \delta, \\ \delta|e| - \frac{1}{2}\delta^2 & |e| > \delta. \end{cases}$$

Gradient Saturation — Preventing Explosion

The gradient $\psi(e)$ clips large residuals to a fixed magnitude δ :

$$\psi(e) = \begin{cases} e & |e| \leq \delta, \\ \delta \cdot \text{sgn}(e) & |e| > \delta. \end{cases}$$

Massive outliers are therefore **bounded** during backpropagation, ensuring stable convergence.

Simulation Setup

OFDM System Parameters

- Subcarriers: $N = 128$
- Cyclic prefix: $N_{cp} = 16$
- Modulation: 64-QAM
- Channel taps: $L = 3$
(freq.-selective)
- Normalized CFO: $\epsilon \sim \mathcal{U}(-0.5, 0.5)$

Training Configuration

- Noise: $S\alpha S$, $\alpha = 1.4$, $\gamma = 0.1$
- Dataset: 20,000 OFDM symbols
- Split: 70% train / 30% validation
- Optimizer: Adam, $LR = 10^{-3}$
- Batch size: 1000; max epochs: 2000

Baseline

All results compare **CAD-HuberNet** against the existing **CAD-MMSE** model [?].
Performance metric: MSE (dB scale, $10 \log_{10}(\text{MSE})$).

Results: Convergence and SNR Performance

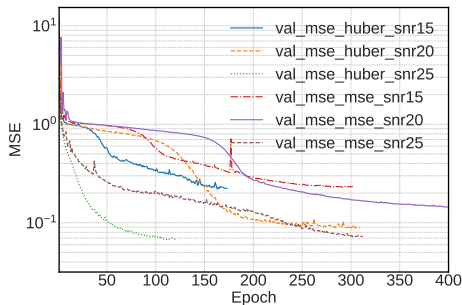


Figure: Training Convergence (MSE vs. Epoch)

MMSE loss oscillates due to impulsive spikes amplifying the squared-error gradient. Huber loss yields **stable, monotone** descent.

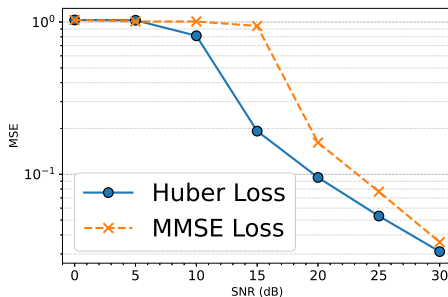


Figure: Steady-State MSE vs. SNR

At 15 dB SNR, CAD-HuberNet achieves **−19.21 dB MSE** vs. **−9.42 dB** for CAD-MMSE — a gain of nearly **10 dB**.

Results: Robustness to Impulsiveness (α)

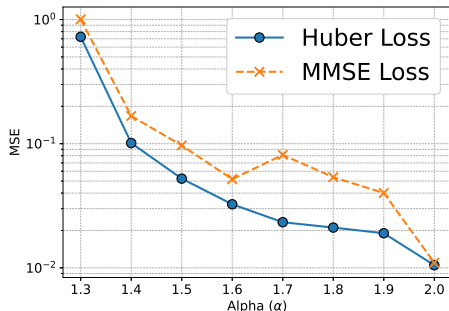


Figure: Steady-state MSE vs. α at SNR = 20 dB

- In the **highly impulsive** regime ($\alpha = 1.3$), CAD-HuberNet achieves MSE ≈ -1.38 dB vs. CAD-MMSE at MSE ≈ 0 dB.
- As $\alpha \rightarrow 2.0$ (Gaussian limit), both models converge to comparable performance, confirming Huber loss incurs **no penalty** in benign conditions.

Conclusion

Summary of Contributions

- Proposed a **robust blind CFO estimation** framework for heavy-tailed non-Gaussian (impulsive) noise environments.
- Designed a **dual-stream CNN-Attention-DNN (CAD)** architecture that jointly exploits power-spectral and phase features.
- Replaced MMSE with **Huber loss**, actively bounding outlier gradients during backpropagation.
- Demonstrated **superior steady-state accuracy and convergence stability** over standard DL baselines, especially for $\alpha < 2$.

Thank You Questions?

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